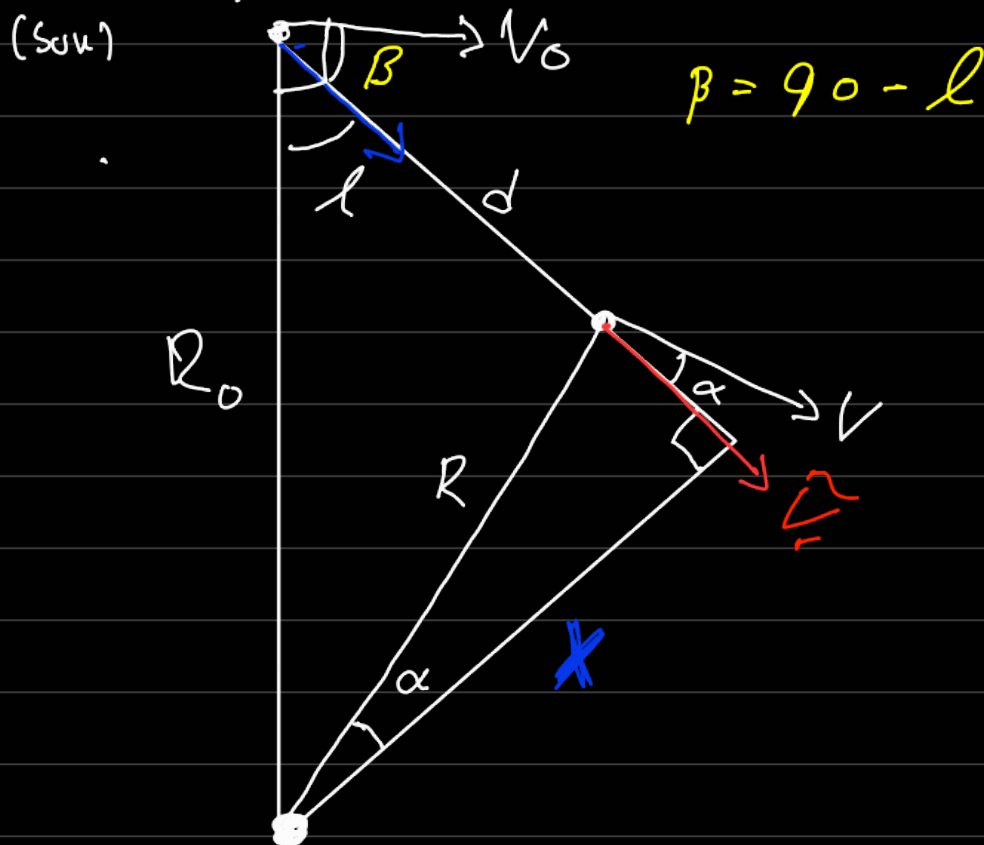


(a) Discussed this already

(b) This exactly what we did in today's lesson. Let's spell it out once more for good measure. Always just remember this geometry:



$$V_r = \cos(\alpha) V - \sin(l) V_0$$

$$\cos\left(\frac{\pi}{2} - l\right) = \cos\left(\frac{\pi}{2}\right)\cos(l) + \sin\left(\frac{\pi}{2}\right)\sin(l) = \sin(l)$$

thus

$$V_r = \cos(\alpha) V - \sin(l) V_0$$

• Use geometry to replace $\cos(\alpha)$:

$$\cos(\alpha) = \frac{x}{R} \quad \text{and} \quad \sin(l) = \frac{x}{R_0}$$

So

$$x = \cos(\alpha) R = \sin(l) R_0$$

or

$$\cos(\alpha) = \frac{R_0 \sin(l)}{R}$$

Plug this in:

$$\begin{aligned} V_r &= \left(\frac{R_0}{R} \right) \sin(l) V - \sin(l) V_0 \\ &= \left(\frac{R_0 V}{R} - V_0 \right) \sin(l) \\ &= \left(\frac{V}{R} - \frac{V_0}{R_0} \right) R_0 \sin(l) \end{aligned}$$

define angular momentum:

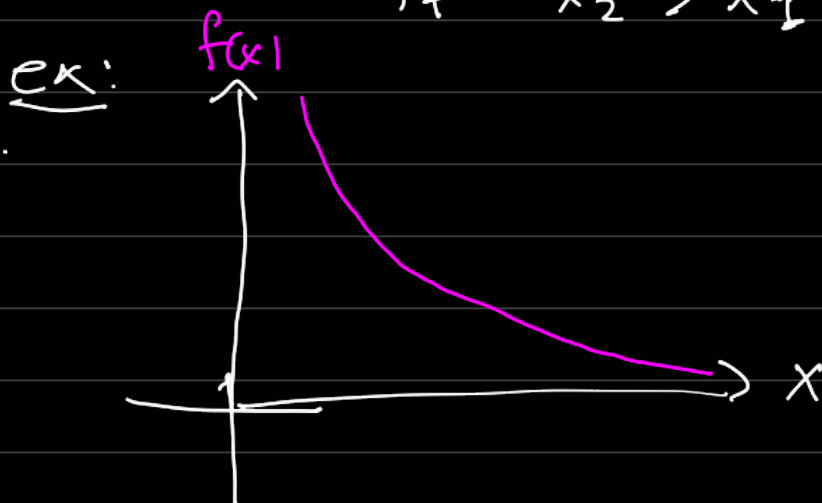
$$\omega = \frac{V}{R} \quad \text{and} \quad \omega_0 = \frac{V_0}{R_0}$$

$$V_r = (\omega - \omega_0) R_0 \sin(l)$$

③ Monotonically decreasing function:

- This is a function which is only decreasing meaning

$$\begin{aligned} f(x_2) &< f(x_1) \\ \text{if } x_2 &> x_1 \end{aligned}$$



for $f(x)$ we see that f is decreasing for all x .

$$f(x) = \frac{1}{x}, \quad x > 0$$

this is an example of a monotonically decreasing f^h

• Note, for $x_1 = 3, x_2 = 4$

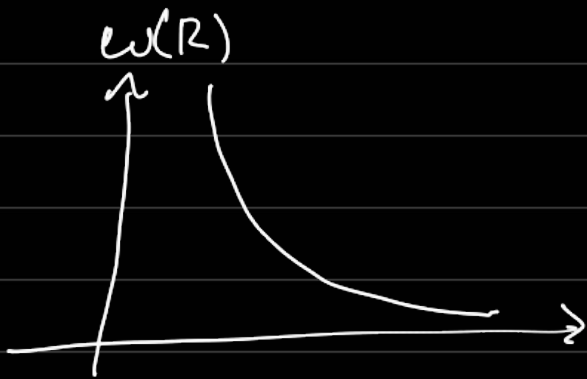
$$f(x_1) = \frac{1}{3}$$

$$f(x_2) = \frac{1}{4}$$

and so

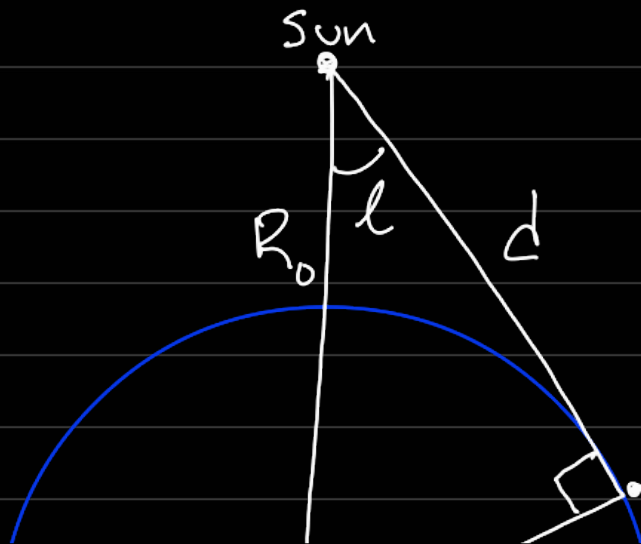
$$f(x_2) < f(x_1)$$

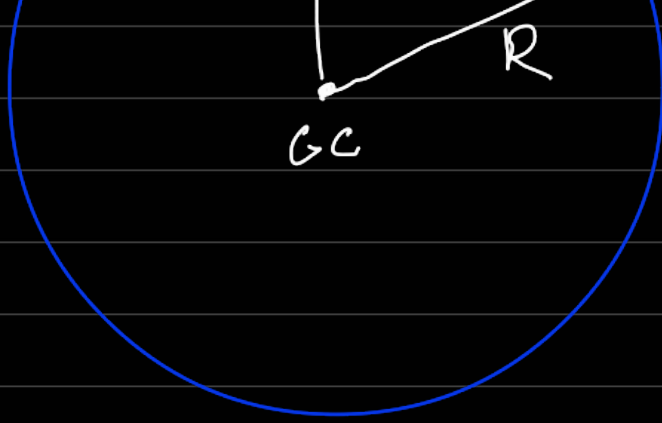
Okay, so that means that w is decreasing for all values of R



Lesson on tangent point method: (pg 29 notes)

• For gas clouds orbiting the Galaxy we can look at HI 21cm lines. These come from





- The tangent point: is the smallest radius along the line of sight from us.

$$\sin(\ell) = \frac{R}{R_0} \Rightarrow \underline{R = R_0 \sin(\ell)}$$

- In the above problem we found

$$V_r = \left(\frac{V}{R} - \frac{V_0}{R_0} \right) R_0 \sin(\ell)$$

we can rewrite this:

$$V_r = \frac{V R_0 \sin(\ell)}{R} - V_0 \sin(\ell)$$

Can replace R

$$\begin{aligned} V_r &= \frac{V R_0 \sin(\ell)}{R_0 \sin(\ell)} - V_0 \sin(\ell) \\ &= V - V_0 \sin(\ell) \end{aligned}$$

and so

$$\boxed{V(R) = V_{r, \text{max}} + V_0 \sin(\ell)}$$

- This is where eq. 20 comes from.

- Because of our assumptions these lines (HI 21cm) are only detected for regions in the inner part of the Galaxy.

- Determine $V^2 = V_r^2 + U_t^2$ from

Don't constants and then use 21cm info
(l, V).

- We will measure $V(R)$ so

$$V(R) = (\omega(R) - \omega_0) R_0 \sin(l) + V_0 \sin(l)$$

Solve for $\omega(R)$

$$\frac{V - V_0 \sin(l)}{R_0 \sin(l)} + \omega_0 = \omega(R)$$

$$R_0 \sin(l) = R$$

$$\left(\frac{V - V_0 \sin(l)}{R} \right) + \omega_0 = \omega(R)$$

$$(\delta) \quad l = 31.5$$

$$V_{\text{rmax}} = 109.9 \text{ km s}^{-1}$$

- Radius: $\frac{R}{\sin(l)} = R_0$

so

$$R = R_0 \sin(l)$$

- Use R_0 from notes: $R_0 \approx 8 \text{ kpc}$

$$R = 8 \text{ kpc} \sin(31.5)$$

- Now, we want to know the orbital velocity:

$$\omega = \frac{V}{R}$$

we solved for R already and we know

$$V = V_{\text{rmax}} + V_0 \sin(l)$$

use v_0 from notes

$$v_0 \approx 218 \text{ km s}^{-1}$$

$$v_{\text{ring}} = 109.9 \text{ km s}^{-1}$$

so

$$\omega = \frac{109.9 + 218 \sin(31.5)}{8 \text{ kpc} \sin(31.5)}$$

